

**Course: Fundamentals of Generative AI and Large Language
Models: Theory and Practice**

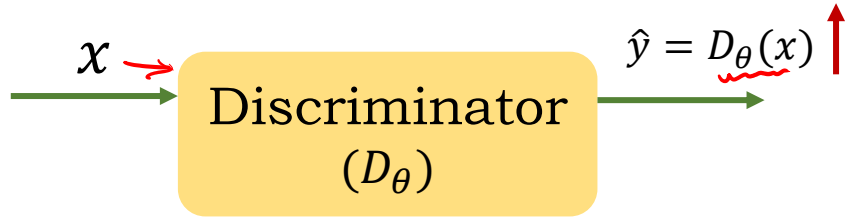
**Week 5: Generative Adversarial Networks (GANs):
Foundations**

Today's Session Overview

GAN Objective and Loss Functions

- Discriminator objective function
- Generator objective function
- GAN objective function

Discriminator loss for a real samples



The training dataset contains real samples drawn from the unknown real-data distribution:

$$x \sim p_{data}(x)$$

Here,

x

represents one real sample from the training dataset.

The sampled real input x is passed to the discriminator:

$$x \rightarrow D_\theta$$

The discriminator produces:

$$\hat{y} = D_\theta(x)$$

where $D_\theta(x)$ is the predicted probability that the input x is real.

Because x is a real training sample, its target label is:

$$y = 1$$

Using BCE:

$$L = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$$

Substituting:

$$y = 1, \quad \hat{y} = D_\theta(x)$$

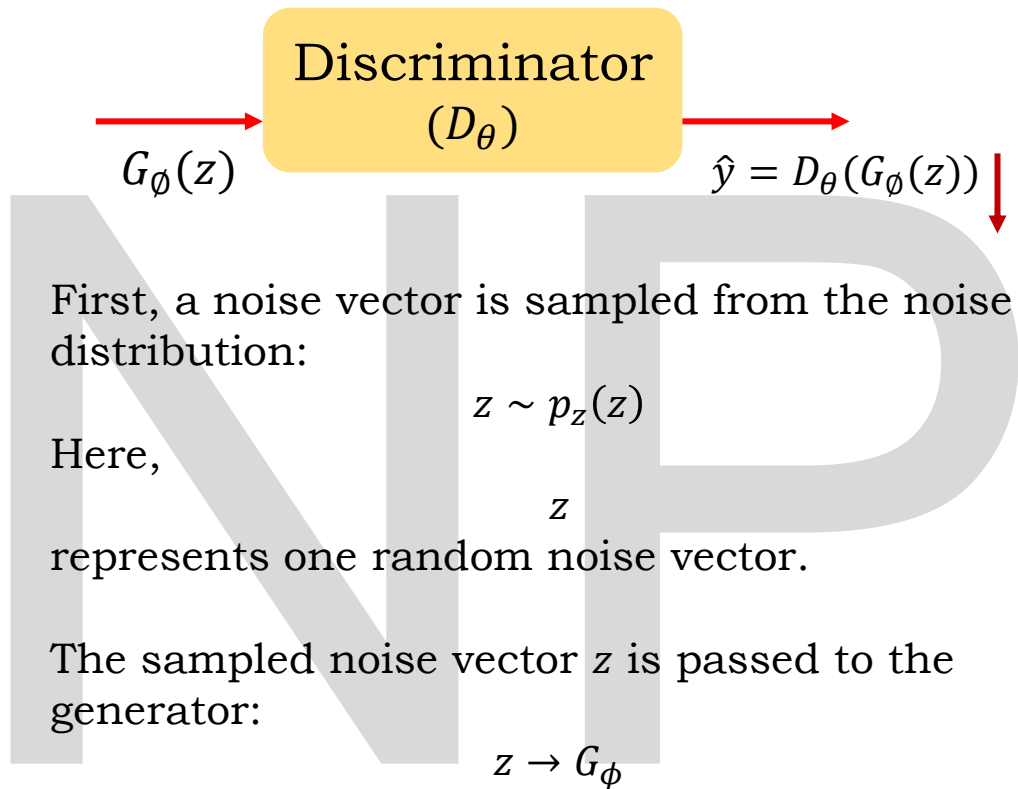
gives:

$$L_{real} = -[1 \log D_\theta(x) + 0 \log(1 - D_\theta(x))]$$

Therefore:

$$L_{real} = -\log D_\theta(x)$$

Discriminator loss for a fake samples



The generator produces a generated sample:

$$x_{fake} = G_\phi(z)$$

The generated sample is then passed to the discriminator:

$$G_\phi(z) \rightarrow D_\theta$$

The discriminator produces:

$$\hat{y} = D_\theta(G_\phi(z))$$

where

$$D_\theta(G_\phi(z))$$

is the predicted probability for the generated sample.

Because $G_\phi(z)$ is a generated sample, its target label for training the discriminator is:

$$y = 0$$

Using binary cross-entropy:

$$L = -[y \log \hat{y} + (1 - y) \log(1 - \hat{y})]$$

Substituting:

$$y = 0, \quad \hat{y} = D_\theta(G_\phi(z))$$

gives:

$$L_{fake} = -\left[0 \log D_\theta(G_\phi(z)) + 1 \log(1 - D_\theta(G_\phi(z)))\right]$$

Therefore:

$$L_{fake} = -\log(1 - D_\theta(G_\phi(z)))$$

Discriminator loss for One Real and One Generated Sample

Combine the discriminator losses

The real and fake losses are combined:

$$L_D = L_{real} + L_{fake}$$

Therefore:

$$L_D = -\log D_\theta(x) - \log(1 - D_\theta(G_\phi(z)))$$

Equivalently:

$$L_D = -\left[\log D_\theta(x) + \log(1 - D_\theta(G_\phi(z)))\right]$$

Neural networks are usually trained by minimizing a loss function. The discriminator parameters θ are updated so that this loss becomes as small as possible. Therefore, we write:

$$\min_{\theta} L_D(\theta; \phi) = \min_{\theta} -\left[\log D_\theta(x) + \log(1 - D_\theta(G_\phi(z)))\right] \leftarrow$$

Here, G_ϕ is kept fixed while the discriminator parameters θ are updated.

$$L_{real} = -\log D_\theta(x)$$

$$L_{fake} = -\log(1 - D_\theta(G_\phi(z)))$$

The discriminator wants to make correct predictions:

$$D_\theta(x) \rightarrow 1$$

for real samples, and

$$D_\theta(G_\phi(z)) \rightarrow 0$$

for generated samples.

Converting loss minimization to objective maximization

$$\min_D [-A(D)] \Leftrightarrow \max_D A(D)$$

Since the discriminator loss is the negative of the objective, minimizing L_D is equivalent to maximizing V_D :

$$L_D = \min_{\theta} - \left[\log D_{\theta}(x) + \log \left(1 - D_{\theta} \left(G_{\phi}(z) \right) \right) \right]$$

is equivalent to:

$$V_D = \max_{\theta} \left[\log D_{\theta}(x) + \log \left(1 - D_{\theta} \left(G_{\phi}(z) \right) \right) \right]$$

This expression uses:

- one real image x ;
- one noise sample z .

During training, we use many real and generated samples. Therefore, we take their average.

For real images:

$$\mathbb{E}_{x \sim p_{\text{data}}} [\log D_{\theta}(x)] \approx \frac{1}{m} \sum_{i=1}^m \log D_{\theta}(x^{(i)})$$

Take many real samples x from the real-data distribution $p_{\text{data}}(x)$, calculate $\log D_{\theta}(x)$ for each sample, and then take the average.

For generated images:

$$\begin{aligned} \mathbb{E}_{z \sim p_z} \left[\log \left(1 - D_{\theta} \left(G_{\phi}(z) \right) \right) \right] \\ \approx \frac{1}{m} \sum_{i=1}^m \log \left(1 - D_{\theta} \left(G_{\phi}(z^{(i)}) \right) \right) \end{aligned}$$

Discriminator objective function

$$\underline{V_D(\theta; \phi)} = \mathbb{E}_{x \sim p_{data}} [\log D_\theta(x)] + \mathbb{E}_{z \sim p_z} \left[\log \left(1 - D_\theta(G_\phi(z)) \right) \right]$$

The semicolon in $V_D(\theta; \phi)$ indicates that θ is being optimized while ϕ is treated as fixed.

Then write:

$$\underline{\max_\theta} \underline{V_D(\theta; \phi)}$$

Understanding why the discriminator maximizes the terms

$$V_D(\theta; \phi) = \mathbb{E}_{x \sim p_{data}} [\log D_\theta(x)] + \mathbb{E}_{z \sim p_z} \left[\log \left(1 - D_\theta(G_\phi(z)) \right) \right]$$

Real-sample term

Consider the real-sample term:

$$\log D_\theta(x)$$

Here,

$$D_\theta(x)$$

is the discriminator's predicted probability that the input x is real.

For a real sample, the desired output is:

$$D_\theta(x) \rightarrow 1$$

At the beginning of training

Initially, the discriminator may not identify the real image correctly. Suppose it gives:

$$D_\theta(x) = 0.1$$

Then:

$$\log D_\theta(x) = \log(0.1) \approx -2.303$$

This is a very low negative value.

Therefore, the discriminator's performance on this real sample is poor.

After some training, suppose:

$$D_\theta(x) = 0.5$$

Then:

$$\log(0.5) \approx -0.693$$

If the discriminator improves further:

$$D_\theta(x) = 0.9$$

then:

$$\log(0.9) \approx -0.105$$

This is closer to 0.

Finally, as:

$$D_\theta(x) \rightarrow 1$$

we get:

$$\log D_\theta(x) \rightarrow \log(1) = 0$$

Therefore:

$$D_\theta(x) \uparrow \Rightarrow \log D_\theta(x) \uparrow$$

The largest limiting value is:

$$\log(1) = 0$$

Hence, maximizing the real-image term pushes:

$$D_\theta(x) \rightarrow 1$$

Understanding why the discriminator maximizes the terms

$$V_D(\theta; \phi) = \mathbb{E}_{x \sim p_{data}} [\log D_\theta(x)] + \mathbb{E}_{z \sim p_z} \left[\log \left(1 - D_\theta(G_\phi(z)) \right) \right]$$

Generated-sample term

Consider the generated-sample term:

$$\log \left(1 - D_\theta(G_\phi(z)) \right)$$

Here,

$$G_\phi(z)$$

is the generated sample, and

$$D_\theta(G_\phi(z))$$

is the discriminator's predicted probability for the generated sample.

For a generated sample, the discriminator wants:

$$D_\theta(G_\phi(z)) \rightarrow 0$$

because the generated sample should be classified as fake.

At the beginning of training

Initially, the discriminator may incorrectly classify. Suppose it gives:

$$D_\theta(G_\phi(z)) = \underline{0.9}$$

Then:

$$1 - D_\theta(G_\phi(z)) = 1 - 0.9 = 0.1$$

Therefore:

$$\log \left(1 - D_\theta(G_\phi(z)) \right) = \log(0.1) \approx \underline{-2.303}$$

This is a very low, highly negative value.

Therefore, the discriminator's performance on this generated sample is poor.

Understanding why the discriminator maximizes the terms

$$V_D(\theta; \phi) = \mathbb{E}_{x \sim p_{data}} [\log D_\theta(x)] + \mathbb{E}_{z \sim p_z} \left[\log \left(1 - D_\theta \left(G_\phi(z) \right) \right) \right]$$

After some training

Suppose the discriminator improves and gives:

$$D_\theta(G_\phi(z)) = 0.5$$

Then:

$$1 - D_\theta(G_\phi(z)) = 1 - 0.5 = 0.5$$

and:

$$\log(1 - D_\theta(G_\phi(z))) = \log(0.5) \approx -0.693$$

If the discriminator improves further

Suppose:

$$D_\theta(G_\phi(z)) = 0.1$$

Then:

$$1 - D_\theta(G_\phi(z)) = 1 - 0.1 = 0.9$$

Therefore:

$$\log(1 - D_\theta(G_\phi(z))) = \log(0.9) \approx -0.105$$

This value is closer to 0.

Finally, as:

$$D_\theta(G_\phi(z)) \rightarrow 0$$

we get:

$$1 - D_\theta(G_\phi(z)) \rightarrow 1$$

and therefore:

$$\log(1 - D_\theta(G_\phi(z))) \rightarrow \log(1) = 0$$

As:

$$D_\theta(G_\phi(z)) \downarrow$$

we have:

$$1 - D_\theta(G_\phi(z)) \uparrow$$

and hence:

$$\log(1 - D_\theta(G_\phi(z))) \uparrow$$

Since the term is $\log(1 - D_\theta(G_\phi(z)))$, its value becomes largest when $D_\theta(G_\phi(z))$ becomes close to zero.

Generator objective function

The GAN value function is:

$$V_G(\phi; \theta) = \mathbb{E}_{x \sim p_{data}} [\log D_\theta(x)] + \mathbb{E}_{z \sim p_z} \left[\log \left(1 - D_\theta(G_\phi(z)) \right) \right]$$

The first term contains only real samples and the discriminator:

$$\mathbb{E}_{x \sim p_{data}} [\log D_\theta(x)]$$

It does not depend on the generator parameters ϕ . Therefore, the generator cannot change this term.

The second term contains $G_\phi(z)$:

$$\mathbb{E}_{z \sim p_z} \left[\log \left(1 - D_\theta(G_\phi(z)) \right) \right]$$

Therefore, during the generator update, we optimize only this term.

The generator wants the discriminator to classify generated samples as real. Therefore:

$$D_\theta(G_\phi(z)) \rightarrow 1$$

Then:

$$1 - D_\theta(G_\phi(z)) \rightarrow 0$$

Therefore:

$$\log \left(1 - D_\theta(G_\phi(z)) \right) \rightarrow -\infty$$

So the generator tries to make this term as small as possible. Hence:

$$\min_{\phi} \mathbb{E}_{z \sim p_z} \left[\log \left(1 - D_\theta(G_\phi(z)) \right) \right]$$

$D_\theta(G_\phi(z))$	$1 - D_\theta(G_\phi(z))$	$\log(1 - D_\theta(G_\phi(z)))$
0.1	0.9	(-0.105)
0.5	0.5	(-0.693)
0.9	0.1	(-2.303)
0.99	0.01	(-4.605)
0.999	0.001	(-6.908)

GAN objective function

The discriminator maximizes with respect to θ :

$$\max_{\theta} V(\theta; \phi)$$

During this update, the generator parameters ϕ are kept fixed.

The generator minimizes with respect to ϕ :

$$\min_{\phi} V(\theta; \phi)$$

During this update, the discriminator parameters θ are kept fixed.

Combining both gives the original GAN minimax objective:

$$\min_{\phi} \max_{\theta} V(\theta; \phi)$$

Therefore:

$$\min_{\phi} \max_{\theta} \mathbb{E}_{x \sim p_{data}} [\log D_{\theta}(x)] + \mathbb{E}_{z \sim p_z} \left[\log \left(1 - D_{\theta}(G_{\phi}(z)) \right) \right]$$

Or

$$\min_G \max_D \mathbb{E}_{x \sim p_{data}} [\log D(x)] + \mathbb{E}_{z \sim p_z} \left[\log \left(1 - D(G(z)) \right) \right]$$

The discriminator wants the generated sample to receive a value close to 0, whereas the generator wants the same generated sample to receive a value close to 1. Therefore, their goals are opposite. Since the discriminator maximizes the GAN value function, the generator minimizes it.

Next Session

- GAN Convergence and Equilibrium

NPTREL

NIPTEEL

Thank you